

TIRTH VAGHELA

AI/ML | Full-Stack Developer

+91 8735028305 | vaghelatirth719@gmail.com | Ahmedabad, Gujarat, India

Portfolio: tirthvaghela.in

GitHub: [Tirthvaghela](https://github.com/Tirthvaghela)

LinkedIn: [tirthvaghela](https://www.linkedin.com/in/tirthvaghela)

CAREER OBJECTIVE

AI/ML & Full-Stack Developer with hands-on experience building computer-vision, NLP, and AI-powered applications using Python, YOLO, OpenCV, OCR, and modern web frameworks. Experienced in developing end-to-end systems that solve real-world problems across AI, automation, and software engineering.

EDUCATIONAL QUALIFICATIONS

Degree	Institution	Duration	CGPA / Status
M.Sc. (IT) — Integrated	GLS University	2025 - 2027	Pursuing (Sem 9)
B.Sc. (IT)	GLS University	2022 – 2025	7.64 CGPA

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, TypeScript, SQL, Dart, PHP

Web Technologies: React.js, Next.js, HTML, CSS, Tailwind CSS, Flutter

Backend / Frameworks: Django, Django REST Framework, Flask, Node.js, Express.js, FastAPI, REST APIs, JWT Auth

Databases: MongoDB, MySQL, SQLite

AI / ML & Computer Vision: YOLOv4, YOLOv8, OpenCV, DeepFace, ArcFace, OCR, BM25

Software / Tools: Git, GitHub, VS Code, Vercel

Operating Systems: Windows

ACADEMIC PROJECTS

VoiceForm AI — AI-Powered Form Filling Assistant | Python, YOLO, OCR, FastAPI, React

- AI-powered document assistant that detects form fields, extracts labels using OCR, and guides users through form completion using multilingual voice interaction. Built the field-detection, OCR, and AI-assisted form-filling pipeline.

ProjectMindAI — Intelligent Project Discovery Platform | Python, Sentence Transformers, BM25, MongoDB

- Intelligent project discovery system combining semantic embeddings and BM25 keyword search for hybrid retrieval, project recommendations, and similar-project discovery. Designed the NLP preprocessing, semantic-search, and hybrid-ranking pipeline.

Urban Signal — AI Traffic Management Platform | Next.js, TypeScript, Flask, YOLOv4, OpenCV, MySQL

- Real-time traffic management system using **YOLOv4 and OpenCV** for vehicle detection, congestion analysis, dynamic signal timing, and emergency-vehicle prioritization. **Built the computer-vision and signal-control pipeline end-to-end.**

RIGS — Roadside Intelligent Governance System | React, Node.js, MongoDB, YOLOv8, OpenCV, OCR

- Automated traffic-violation detection using **YOLOv8 and OCR** to identify helmetless riders, extract license plates, and generate e-challans. **Built the detection, OCR, and challan-generation pipeline.**

Wedding Face Finder — AI Photo Search System | Flask, DeepFace, ArcFace, RetinaFace

- AI-powered facial search system using DeepFace, ArcFace, and RetinaFace to identify users across thousands of photos through facial embeddings and similarity matching. Implemented the embedding and similarity-search pipeline.

INTERNSHIP / WORK EXPERIENCE

Full-Stack Web Development Intern — Prodigy InfoTech (Remote)

February 2026

- Built a production-ready enterprise authentication system (Django REST Framework + React) with JWT, 2FA (TOTP), session management, audit logging, and RBAC.
- Developed a MERN-stack Employee Management System (JWT auth, file uploads, search/filtering, CSV/PDF export) and a PHP & MySQL e-commerce platform with cart, wishlist, orders, and admin dashboard.
- Built Vois, a full-stack social platform (React, FastAPI, MongoDB) with JWT auth, posts, comments, likes, and follow system.

CERTIFICATIONS

- Master MongoDB Database Design (Compass, Mongosh, PyMongo) — Udemy, Feb 2026
- Claude Code in Action — Anthropic, Mar 2026
- Developing Front-End Apps with React — IBM (Coursera, 4 modules)

ADDITIONAL INFORMATION

Soft Skills: Problem Solving, Team Collaboration, Communication, Adaptability, Time Management, Self-Learning

Languages Known: English, Hindi, Gujarati

Hobbies / Interests: Video Editing, Drawing, Guitar Performance, Exploring New Tech, Music While Coding